

financial crisis, in contrast, there was a large contemporaneous increase in volatilities and credit spreads in 2007 and 2008 and then a decrease as markets stabilized post-2009. In summary, *credit investments are short volatility trades*.

4.3. CREDIT SPREAD PUZZLE

The expected loss from default is given by

$$\text{Expected Loss from Default} = \text{Default Probability} \times (1 - \text{Recovery Rate}), \tag{9.5}$$

where the recovery rate is the fraction of the bond’s face value that can be recovered by the bondholders when the firm defaults. Structural models of default give guidance for various factors that should affect default and recovery rates, most notably volatility, as discussed above. Although credit spreads do reflect default risk, it turns out that a significant fraction of movements in credit spreads is unrelated to variables measuring default and recovery rates. Furthermore, corporate spreads are several times wider than those implied by just expected default losses in simple structural models. This is the *credit spread puzzle*.²⁷

Table 9.15 reports credit spreads above Treasury yields compared to credit loss rates from Moody’s. As in equation (9.5), credit loss rates include losses from default net of any sums recovered. The spread is many times higher than the credit loss rates for investment grades—several orders of magnitude higher in fact. However, the credit loss rates are approximately the same as yields for junk bonds.

The credit spread puzzle is observed over much longer samples than the one reported in Table 9.15. Giesecke et al. (2011) study U.S. corporate bond spreads over 150 years, from 1866 to 2008. They find long-run average credit losses are around 0.75% per year, but there are also pronounced waves of much higher default losses: the 1870s (railroad boom and crash), the 1930s (Great Depression),

Table 9.15

Corporate Spreads and Credit Losses (%)					
	Aaa	Aa	A	Baa	Caa
Corp Spreads vs Trsys 1987–2011	6.20	6.35	6.72	88.20	15.40
Credit Loss Rates 1982–2008 (Moody’s)	0.00	0.03	0.02	0.12	13.34
Credit Loss = Default and Recovery					

²⁷ See Huang and Huang (2012) for a summary of the credit spread puzzle.